

Title

Temporal Study 2021-2022: Sample-Based Surface Water Dissolved Inorganic Carbon, Dissolved Organic Carbon, Total Nitrogen, Total Suspended Solids, Ions, and Organic Matter from across Multiple Watersheds in the Yakima River Basin, Washington, USA (v3)

Summary

This dataset supports a broader study examining the drivers of temporal variability in sediment respiration rates in the Yakima River Basin. The dataset provides geochemistry and organic matter characterization data generated from samples collected at weekly or bi-weekly intervals at six sites across the Yakima River Basin in Washington, USA. Related sensor data will be published separately and can be used to link sediment respiration rates to biogeochemical processing rates.

The data package was originally published in November 2022. It was updated in November 2023 (v2; modified files) and April 2025 (new and modified files). See the change history section below for more details.

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

Brief Overview of Methods

Surface water samples were collected at six sites by 2-person teams within the American River, Little Naches River, and Yakima River in late April 2021 through early April 2022. Site metadata, including general environmental information, were manually recorded in the field. For complete information regarding data collection, see the Sampling Protocol for details on water sample and site metadata collection. Surface water samples were processed and analyzed in the laboratory after field collection. For details regarding laboratory methods, see the alphanumeric methods codes located in the header rows of the chemistry data csv file and their associated definitions in RC2_Methods_Codes_2021-2022.csv. A separate dataset will be published with data generated from the sensors deployed in the field and grain size photographs.

Critical Details

Name	Parent_ID	Sample_Name	Sample_Name
Example	RC2_0350	RC2_0350_Water	RC2_0350_TSS-1 RC2_0350_OCN-1 RC2_0350_OCN-2 RC2_0350_OCN-3
Description	All samples from one sampling event start with the Parent_ID . Parent_ID can be found in RC2_Sample_Field_Metadata.csv.	When referring to all samples of a given material from one sampling event, the Sample_Name consists of the Parent_ID followed by an indicator of the material. An example of this can be found in RC2_Sample_IGSN-Mapping.csv.	When referring to individual samples, the Sample_Name consists of the Parent_ID followed by an analysis-specific indicator code and may include the replicate number. An example of this can be found in RC2_TSS_2021-2022.csv and RC2_NPOC_TN_2021-2022.csv *The sample types and example files listed are non-exhaustive

- 1 – Each sampling event has a unique Parent_ID in the format RC2_#. The field metadata and data files all contain these unique IDs and can be mapped across each other accordingly. The Parent_ID may have other indicators appended when referring to samples. See figure above for details.
- 2 – The Site_ID can be used to determine the sampling for each site. The Parent_ID for the different sampling events at the same site is unique. The Field Metadata contains the Site_ID and Parent_ID and can be used to map the Site_ID for each sample.

Data Package Structure

This dataset is comprised of one main data folder containing (1) file-level metadata; (2) data dictionary; (3) field metadata; (4) dissolved inorganic carbon (DIC), dissolved organic carbon (DOC; reported as non-purgeable organic carbon; NPOC), total nitrogen (TN), total suspended solids (TSS), and ions; (5) averaged values from water chemistry data; (6) surface water sampling protocol; (7) sensor protocol; (8) readme; (9) methods codes; (10) international generic sample number (IGSN) mapping file; and (11) folder of high resolution characterization of organic via 12 Tesla Fourier transform ion cyclotron resonance mass spectrometry (FTICR-MS) through the Environmental Molecular Sciences Laboratory (EMSL; <https://www.pnnl.gov/environmental-molecular-sciences-laboratory>). This folder contains two subfolders, one containing the .xml data files and the other containing instructions for using Formularity (<https://omics.pnl.gov/software/formularity>) and an R script to process the data based on the user's specific needs. All files are .csv, .pdf, .R, .ref, or .xml.

Acknowledgements

We acknowledge the Yakama Nation as owners and caretakers of the lands where we collected our data. We thank the Confederated Tribes and Bands of the Yakama Nation Tribal Council and Yakama Nation Fisheries for working with us to facilitate sample collection and optimization of data usage according to their values and worldview.

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Change History

Approach to change history and versioning:

Updates to **data package** version: When any file within a data package is updated, the data package version number is updated. The data package version number is indicated in the title of the data package, the data package folder name, and in the change history table below. You can access previous versions of the data package by sending a request to ESS-DIVE.

Updates to **individual file** versions: As files are changed, the file version number is also updated. The file version number is indicated in the file name, file level metadata (flmd) file, and the change history table

below. The version number on an individual file may not match the version number of the data package. For example, v3 of a data package may include v2 of an individual file.

The change history below describes each file revised during versioning. If you are interested in seeing the exact cells within a file that have changed, you can utilize the daff package in R (<https://github.com/edwindj/daff>) to compare a previously downloaded file to a newly downloaded file.

In the change history table below, the sub-headers and bullets indicate the type of change in each file:

- New files: Describes new files added that were not present in previous data package versions
- Bulk changes to files: Describes a change to many files within the data package. The indicated superscript with be added to each file name that the change applies to.
- Modified files:
 - Corrected: Describes existing information modified or removed to prevent sharing of incorrect information
 - Added: Describes new information inserted into an existing file (e.g., appending new columns/rows)
 - Updated: Describes modifying existing information to maintain accuracy though version changes. (e.g., changing version number to new version number)

Change history:

Data Package Version	Changes
Version 1 <i>November 2022</i>	Original data package publication
Version 2 <i>November 2023</i>	<u>MODIFIED FILES</u> RC2_Sample_Field_Metadata.csv (v2) - Corrected year in "Date" from 2021 to 2022 for Sample IDs RC2_0177 through RC2_0211 RC2_Sample_IGSN-Mapping.csv (v2) - Corrected year in "Collection_Date" from 2021 to 2022 for Sample IDs RC2_0177 through RC2_0211 - Added the "10.58052/" DOI prefix to IGSNs and Parent_IGSNs. - Updated IGSN acronym definition to align with current guidelines. readme_RC2_Sample.pdf (v2) - Added new versioning information. - Updated version number in data package title. RC2_Sample_flmd.csv (v2) - Updated version number to corrected, added, and/or updated files. - Update date column headers to align with current guidelines. - Updated IGSN acronym definition to align with current guidelines. RC2_Sample_dd.csv (v2) - Updated IGSN acronym definition to align with current guidelines.
Version 3 <i>April 2025</i>	<u>BULK CHANGES TO FILES</u> ¹ indicates international generic sample number (IGSN) was added to the file

	² indicates the data was corrected to current QAQC approach which might include: remove blank correction, add additional samples, update text flags, update methods deviations <u>MODIFIED FILES</u> RC2_NPOC_TN_2021-2022.csv (v2) ^{1, 2} - Previously separated into “RC2_NPOC_2021-2022.csv” and “RC2_TN_2021-2022.csv” RC2_Ions_2021-2022.csv (v2) ^{1, 2} - Removed NH4, Br, F, Li, Mg, K, Na, Ca due to data quality concerns - Note: Ions were never blank corrected RC2_Sample_Data_Summary_2021-2022.csv (v2) ¹ - Updated to reflect changes in analyte files RC2_Methods_Codes_2021-2022.csv (v2) - Corrected description to remove indication of original QAQC approach (blank correction) RC2_Sample_Field_Metadata.csv (v3) - Updated “Sample_Name” column to “Parent_ID” - Removed “GPS_Accuracy” column readme_RC2_Sample.pdf (v3) - Updated version number in title - Added new versioning information. RC2_Sample_flmd.csv (v3) - Updated version number and file names of modified files. RC2_Sample_dd.csv (v3) - Removed rows to reflect removed ion columns RC2_TSS_2021-2022.csv (v2) ^{1, 2} RC2_FTICR_2021-2022.csv (v2) ¹ RC2_DIC_2021-2022.csv (v2) ^{1, 2}
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Note: An asterisks (*) in a file name indicates multiple files with the same file and file naming structure. See the flmd for more details about specific files and the meaning of the asterisks.